

ROTVAER, Norway

WMO: 011050

Lat: 68.368N

Lon: 15.943E

Elev: 26

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.1	19.9	7.8	8.2	19.7	10.3	9.3	21.8	50.9	40.8	45.3	37.7	20.0	70	0.893

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	67.9	58.2	64.5	56.8	61.9	55.3	59.4	65.4	57.7	63.0	56.2	60.9	8.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.3	67.5	62.3	54.9	64.2	60.7	53.5	60.9	58.9	26.0	65.4	24.9	63.2	23.9	61.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
42.5	36.5	31.8	14.4	75.4	5.0	5.9	10.7	79.7	7.8	83.1	5.0	86.5	1.3	90.8
			12.8	61.4	4.2	1.8	9.8	62.7	7.3	63.8	5.0	64.8	1.9	66.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	30.8	30.1	32.0	37.0	44.2	49.8	55.6	55.5	51.0	43.0	37.9	34.4
	DBStd	10.33	5.93	6.14	4.91	4.14	5.38	4.88	4.86	4.02	4.01	4.84	5.08	5.16
	HDD50	3467	596	558	559	389	195	60	2	3	35	221	364	485
	HDD65	8460	1061	978	1024	839	644	456	296	296	421	681	814	950
	CDD50	488	0	0	0	0	16	54	176	173	64	5	0	0
	CDD65	7	0	0	0	0	0	1	5	1	0	0	0	0
	CDH74	18	0	0	0	0	2	2	12	2	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	15.6	17.7	17.6	17.0	14.7	14.5	14.4	13.8	13.6	13.9	15.7	16.3	17.8
Precipitation	PrecAvg	60.7	6.3	5.8	5.9	3.6	2.9	2.6	3.3	4.2	5.3	7.5	6.4	6.8
	PrecMax	76.7	10.5	9.8	9.1	6.4	5.2	5.0	6.3	9.8	11.0	15.0	11.9	12.4
	PrecMin	51.9	2.1	2.9	1.1	1.3	0.9	0.9	1.1	1.8	2.4	2.0	1.9	3.3
	PrecStd	6.6	2.4	2.0	2.2	1.6	1.2	1.2	1.7	1.7	2.1	3.1	3.4	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	43.1	44.1	50.9	66.5	69.5	74.8	69.6	62.4	56.3	49.6	46.8
		MCWB	40.5	40.0	38.9	43.1	53.6	56.2	63.3	59.0	53.9	49.4	46.7	42.8
	2%	DB	42.9	41.7	41.9	47.2	58.1	63.4	69.3	65.6	60.0	53.6	47.9	45.1
		MCWB	39.8	39.0	38.3	41.5	49.3	52.9	59.6	57.8	52.6	48.3	44.4	42.0
	5%	DB	41.0	40.4	40.7	43.9	54.3	59.3	65.7	63.2	58.0	51.4	46.8	43.7
		MCWB	38.6	38.0	37.6	39.5	47.3	51.6	58.1	56.8	52.0	47.8	43.1	40.4
	10%	DB	39.0	38.7	39.4	42.0	51.0	56.4	62.6	61.1	55.8	49.5	45.2	41.8
		MCWB	36.3	36.2	36.8	38.4	45.2	50.5	56.4	55.7	51.4	46.2	41.6	38.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	41.4	40.4	44.0	54.3	57.0	64.2	60.9	55.1	50.7	47.3	44.7
		MCDB	43.4	42.4	42.3	50.1	67.5	67.6	71.6	65.7	59.5	53.3	48.6	45.6
	2%	WB	40.2	39.5	39.2	41.9	50.0	54.6	60.8	58.9	53.9	49.7	45.6	42.8
		MCDB	42.1	40.9	41.1	45.6	57.1	60.8	68.0	64.2	57.9	52.5	47.5	45.0
	5%	WB	38.6	38.2	38.0	40.5	47.4	52.6	58.6	57.4	52.9	48.1	43.7	40.4
		MCDB	40.9	40.1	40.3	42.9	53.5	57.9	64.3	62.1	56.5	51.2	46.3	43.3
	10%	WB	36.5	36.4	36.8	39.3	45.7	51.0	56.9	56.0	51.9	46.4	41.7	38.5
		MCDB	38.8	38.6	39.3	41.6	50.2	55.7	61.8	60.3	55.2	49.6	44.8	41.3
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.4	5.6	6.0	7.1	6.9	7.6	6.7	5.4	4.6	5.0	5.3
		MCDBR	6.6	7.0	6.2	8.0	13.6	12.8	14.1	11.0	8.1	5.6	6.3	7.7
		MCWBR	6.0	6.5	5.1	4.8	6.1	5.9	6.1	5.0	4.1	4.4	5.6	7.1
	5% WB	MCDBR	6.4	6.8	6.5	6.9	12.4	11.0	13.4	10.0	6.5	5.1	6.3	7.8
		MCWBR	6.2	6.6	5.8	5.0	5.9	5.8	6.4	5.0	4.0	5.0	6.3	7.6
Clear-Sky Solar Irradiance	taub	0.372	0.272	0.310	0.331	0.340	0.341	0.346	0.336	0.319	0.283	0.352	N/A	
	taud	2.647	2.117	2.206	2.318	2.429	2.496	2.511	2.528	2.543	2.276	2.609	N/A	
	Ebn at Noon	36	159	226	259	270	273	268	258	232	157	31	0	
	Edh at Noon	4	15	24	30	30	29	27	24	18	12	3	0	
All-Sky Solar Radiation	RadAvg	8	102	455	1013	1459	1544	1409	1020	540	183	23	0	
	RadStd	1	23	70	87	118	216	153	101	55	34	2	0	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page